

Biomedical Research Assistant

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Biomedical Research Assistant — AI-Powered Literature Research

Helping biomedical researchers move from information overload to evidence-grounded insight faster.



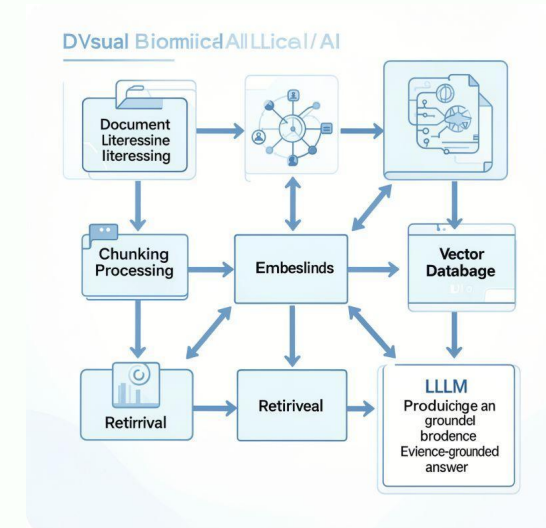
The Challenge

- Biomedical research means massive, fast-growing literature.
- Manual review is time-consuming and inefficient.
- Researchers need faster, context-aware access to domain knowledge.



The Solution

- **MEL-AI**: a specialized RAG assistant for melanoma research with a curated knowledge base.
- **BioLit-AI**: a custom literature assistant for PDFs or PMCIDs, designed for question answering.



How It Works

Scientific Literature → Document Processing → Chunking → Embeddings → Vector Database → Retrieval → LLM → Evidence-grounded Answer

Knowledge Base at a Glance

Data processing pipelines for MEL-AI and BioLit-AI.

Literature Pipeline

5,000

Downloaded
(PubMed)

4,783

After Cleaning
(- 217 dropped)

15,315

Chunks in Pinecone
(embedded & indexed)

Abstracts cleaned, deduplicated, and chunked before embedding.

Drug Pipeline

100

ChEMBL IDs
Extracted

57

Used After Cleaning
Approved compounds

113

Chunks in Pinecone
Processed & indexed

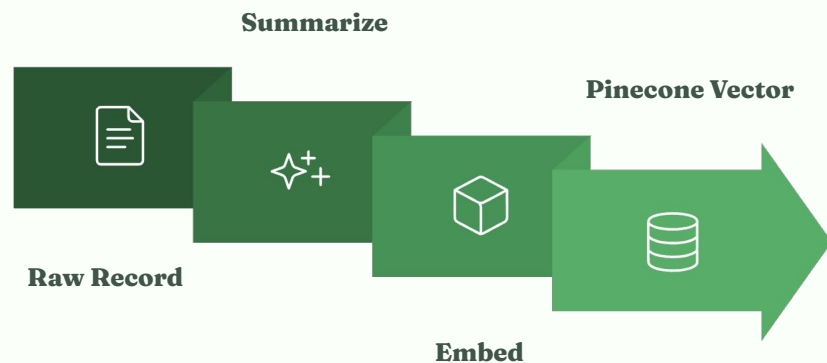
Sourced from ChEMBL (mechanism, targets, indications) + PubChem (descriptions), merged per drug.

Total Knowledge Base: 15,428 chunk across 2 namespaces

How It Works: Smart Embedding & Routing

Embedding & Citation Strategy

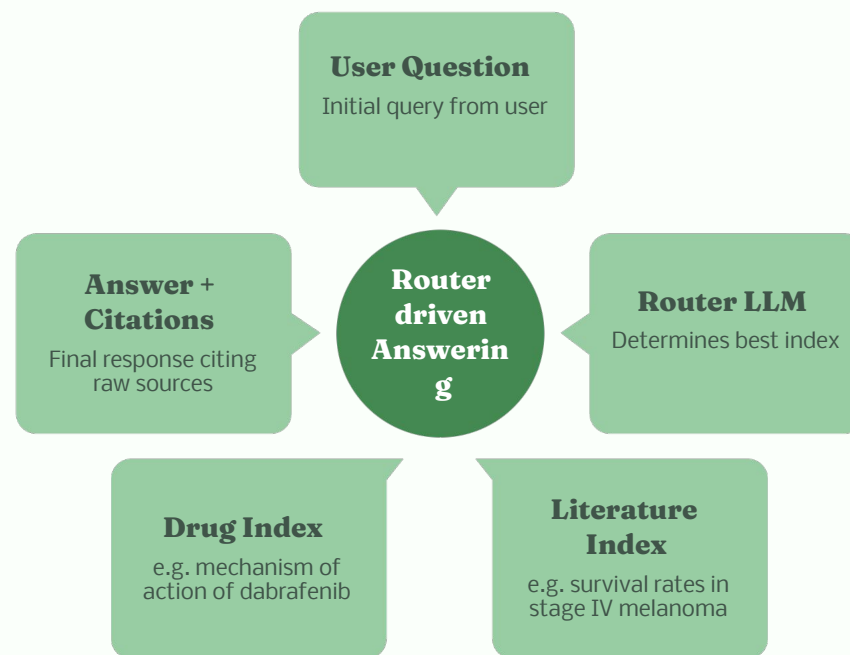
Raw records processed for semantic search while preserving verifiable citations



- ❏ Answer cites the raw source – never the summary – so citations stay verifiable.

Query Routing

Questions intelligently routed to the most relevant knowledge base



The Router LLM classifies each question – whether it pertains to literature, drugs, or both – before any retrieval occurs.

BioLit-AI — Ask Questions Across Biomedical Papers

Multimodal RAG assistant for biomedical literature review

Key Features

-  Ingest up to 10 papers – upload PDFs or fetch directly by PMCID
-  Parses text, tables, figures & section structure (PDF); fast text-only mode for PMCID
-  Retrieves relevant content and answers in natural language
-  Every answer cited – document, section/subsection, page or PMC link
-  Built as a lightweight MVP: Chroma vector store + LangChain MultiVectorRetriever

Process Flow



Disclaimer: Not a diagnostic tool – answers only from ingested documents.

Development Challenges & Current Limitations

Major Development Challenges

1. LangChain Compatibility

- Frequent API changes across versions
- Version conflicts with core dependencies
- Required extensive debugging and pinning

2. PDF Extraction & Cleaning

- Scientific PDFs have complex layouts
- Cleaning and chunking were essential
- Processing varied by article format

Current Limitations

1. Limited Domain Knowledge

- Focused melanoma knowledge base
- Drug coverage remains selective

2. Slow Ingestion

- PDF/PMCID parsing takes time
- High-resolution parsing is costly
- Workflow is still end-to-end sequential

 **Key Takeaway:** The MVP proves the RAG workflow, but speed, scale, and coverage are the next priorities.

LIVE DEMO LINK: <https://biomedical-literature-assistant.streamlit.app/>